

(b) Crude oil	
Students should:	
4.7	know that crude oil is a mixture of hydrocarbons
4.8	describe how the industrial process of fractional distillation separates crude oil into fractions
4.9	know the names and uses of the main fractions obtained from crude oil: refinery gases, gasoline, kerosene, diesel, fuel oil and bitumen
4.10	know the trend in colour, boiling point and viscosity of the main fractions
4.11	know that a fuel is a substance that, when burned, releases heat energy
4.12	know the possible products of complete and incomplete combustion of hydrocarbons with oxygen in the air

Students should:	
4.13	understand why carbon monoxide is poisonous, in terms of its effect on the capacity of blood to transport oxygen <i>references to haemoglobin are not required</i>
4.14	know that, in car engines, the temperature reached is high enough to allow nitrogen and oxygen from air to react, forming oxides of nitrogen
4.15	explain how the combustion of some impurities in hydrocarbon fuels results in the formation of sulfur dioxide
4.16	understand how sulfur dioxide and oxides of nitrogen contribute to acid rain
4.17	describe how long-chain alkanes are converted to alkenes and shorter-chain alkanes by catalytic cracking (using silica or alumina as the catalyst and a temperature in the range of 600–700 °C)
4.18	explain why cracking is necessary, in terms of the balance between supply and demand for different fractions